Turnout And Vote-Share Residuals: A Method Framework For Investigative Election Forensic Reports

Gio Della-Libera

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Abstract

Forensic analytics can prioritize investigation. They do not prove fraud, certify outcomes, or replace risk-limiting audits. This paper applies that W-series boundary to two residual families: turnout residuals and candidate vote-share residuals. A residual is the difference between an observed value and a model-predicted value for a reporting unit, where the prediction is fitted from comparable units, history, demographics, or other public features. The method is deliberately modest: fit a baseline, compute standardized or studentized residuals, rank units by magnitude, and attach caveats explaining why an outlier may be innocent. The paper instantiates the W.01 forensic report contract for residual methods with method identity, feature set, baseline population, model identity, unit scores, calibration by rank or p-value, and an investigation note, all bound to source package hashes. It is a method and framework paper only. No forensics crate, fixture suite, or shipped test is claimed here.

1 Introduction

Forensic analytics can prioritize investigation. They do not prove fraud, certify outcomes, or replace risk-limiting audits. That boundary comes first because residual methods are easy to overread: a large residual can be a useful triage signal, but it is not a finding about intent, legality, or correctness.

This paper covers two W-series analytic families. Turnout residuals ask whether a reporting unit's turnout is unusual after comparison with its own history, nearby or similar units, and public covariates such as population composition or registration patterns. Candidate-share residuals ask whether a contest share is unusual relative to comparable units, previous contests, or cast vote record features where those records are available. In both cases, a residual means observed minus model-predicted: the reported value for a unit less the value a baseline model would have expected from the comparison population.

The separation from the V-series is structural. V-series work verifies canvass and audit claims under explicit evidence contracts. W-series residual reports produce an investigation queue. They may help an analyst decide which precinct, batch, scanner, or contest deserves review, but they cannot certify the election and must not be stored or read as certification evidence. The W.01 overview fixes that claim boundary and the common forensic report contract [Della-Libera, 2026b]; this paper instantiates the contract for turnout and vote-share residuals.

2 Method

The residual method has four steps. First, choose a baseline population: prior elections for the same unit, comparable reporting units in the current election, neighboring units, demographic or registration covariates, or a combination of those sources. Second, fit a baseline model that predicts turnout or candidate share for each unit. Third, compute the residual

$$r_i = y_i - \widehat{y}_i,$$

where y_i is the observed turnout rate or vote share for unit i and $\widehat{y}_i$ is the model prediction. Fourth, standardize the residuals and rank units by magnitude.

The statistic can be reported as a standardized residual, a studentized residual, a robust score, or a rank within the baseline population. A simple implementation might divide by an estimated residual standard deviation; a more careful one can use robust regression, cross-validation, heavy-tailed errors, or leave-one-out studentization so that one extreme unit does not define the baseline that judges it. The report should identify the model as a residual model and name the feature set rather than present the score as self-evident.

A unit can be a residual outlier for innocent reasons. New housing, a college dormitory, a changed precinct boundary, a local uncontested race, language access changes, weather, ballot style, or a mobilization campaign can all move turnout or vote share away from what history predicted. The method therefore ranks units for review; it does not explain the residual by itself.

3 RCOUNT Inputs And Report Contract

Residual reports use RCOUNT records only as normalized, hash-bound inputs. For turnout residuals, the primary inputs are reporting-unit summaries and precinct lineage: registered or eligible population where available, ballots cast, units that split or merged, and the mapping needed to compare a unit with its history. For vote-share residuals, the primary inputs are summaries and, where public and normalized, cast vote records. Summaries supply contest totals and candidate shares; CVRs can support ballot-style, batch, or scanner covariates without turning the residual score into certification evidence.

The W.01 contract defines the shared forensic report shape [Della-Libera, 2026b]. For this method, the per-record fields are instantiated as follows.

Field	Residual-method meaning
`method_id`	Turnout residual or vote-share residual analytic.
`feature_set`	History, demographic covariates, comparable-unit shares, CVR-derived covariates, or other public inputs used by the model.
`baseline_population`	Units and time window used for comparison.
`model_id`	`residual`.
`unit_id`	Reporting unit, contest, batch, or scanner-associated unit scored by the model.
`score`	Standardized residual, studentized residual, robust residual score, or absolute residual rank statistic.
`p_value_or_rank`	Optional p-value, empirical tail probability, or rank in the investigation queue.
`investigation_note`	Machine-readable caveat, including baseline limits or known benign explanations.

The report must also carry the source package hashes from which the features were derived. Hash binding lets another analyst rerun the same registered method against the same RCOUNT package and determine whether the same residual queue is obtained. This paper defines the intended contract use for residuals; it does not claim that a crate, function, report fixture, or test currently implements it.

4 Fixture Pattern

The residual family should follow the four fixture roles defined for W-series analytics. These are intended fixture patterns, not claims about shipped tests. They describe what later implementation work should exercise.

Fixture	Residual-method role
No-anomaly	A synthetic package whose turnout and candidate shares are drawn from the fitted baseline. A residual method should return low scores or an empty low-priority queue.
Injected outlier precinct	A package with one planted reporting unit whose turnout or candidate share is shifted away from comparable units. The method should surface that unit near the top of the queue.
Batch/scanner-correlated residual	A package where several high residuals share a batch or scanner grouping. The report should preserve the grouping signal instead of treating each unit as unrelated.
False-positive boundary	A package where an unusual residual has a benign explanation, such as a new development, a college dormitory, or a changed reporting-unit boundary. The investigation note should prevent the score from being read as proof.

The false-positive boundary is central. A residual method succeeds only if it can rank unusual units while carrying enough context to show that unusual is not the same as fraudulent, incorrect, or certifying.

5 Boundaries

Forensic analytics can prioritize investigation. They cannot prove fraud, certify correctness, or replace risk-limiting audits. Residual methods require that sentence because they produce familiar-looking statistics whose limits are easy to forget.

The main technical boundary is baseline quality. A sparse, biased, stale, or misaligned baseline can manufacture false outliers. If a precinct changed shape, if comparable units are chosen poorly, if demographic covariates omit a new population center, or if historical turnout came from a different contest type, then the residual is partly measuring the baseline defect. The report must name the feature set and baseline population so that this weakness is visible.

The second boundary is causal. A residual is descriptive: it says that an observed turnout rate or vote share was far from what the model predicted. It does not say why. The explanation may be administrative, demographic, political, geographic, or data-quality related. Human investigation and, where appropriate, audit procedures remain outside the residual statistic.

6 Conclusion

Turnout and vote-share residuals are useful W-series methods because they turn normalized RCOUNT records into a reproducible investigation queue. The method is simple in outline: define comparable inputs, fit a baseline, compute observed minus predicted values, standardize the residuals, and rank units with caveats attached.

The discipline is more important than the statistic. Residual reports must bind to source package hashes, identify their feature set and baseline population, use the W.01 forensic report contract, and carry investigation notes that keep outliers from becoming claims. This paper is therefore a method framework, not an implementation announcement: it specifies how residual analytics should be reported when built, while preserving the W-series rule that anomaly scores are investigative, never proof and never certification.

References

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